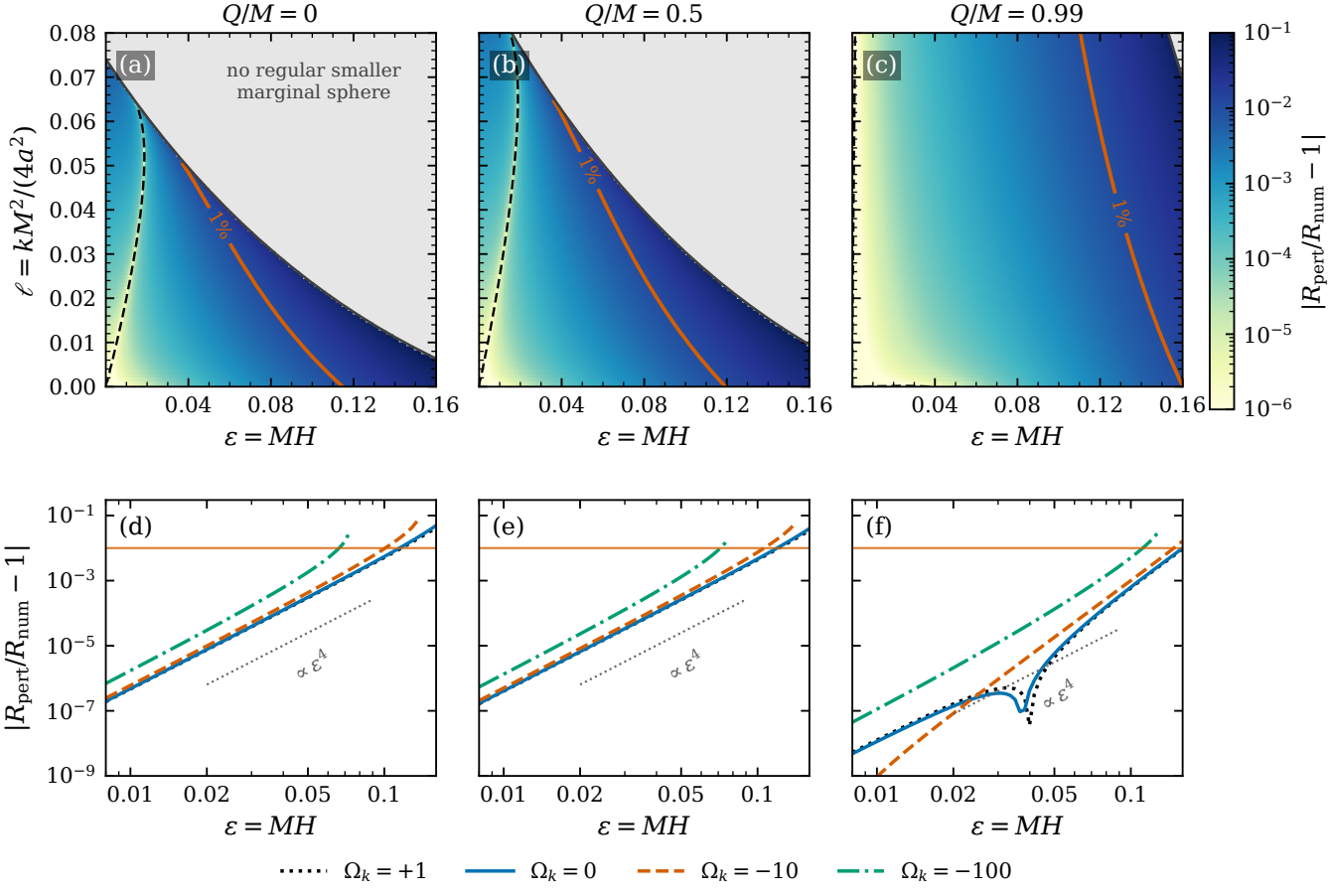

Q/M = 0
Q/M = 0.5
Q/M = 0.99
(a)
(b)
(c)
no regular smaller
marginal sphere
1%
$\ell = kM^2/(4a^2)$
$\varepsilon = MH$
$|R_{\rm pert}/R_{\rm num} - 1|$
(d)
(e)
(f)
$\propto \varepsilon^4$
$\Omega_k = +1$
$\Omega_k = 0$
$\Omega_k = -10$
$\Omega_k = -100$